

Exact Ground Truth for Weld-Seam Extraction: A Constructed Benchmark and a Controlled Comparison of Seven Published Methods

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Abstract—Robotic welding needs the three-dimensional curve along which two parts meet, and a growing body of 3D-vision work extracts that seam from point clouds with reported sub-millimetre accuracy. That accuracy is measured against hand-taught paths or hand-clicked points, on a few workpieces under one set of conditions, with data that is rarely shared; a common benchmark does not exist. We present one in which the ground truth is *constructed, never detected*: scenes of parts waiting to be welded are generated from the seam outward, so every seam is known analytically, every condition (joint type, curvature, groove, thickness, fixture, viewpoint, sensor noise) is a controlled variable, and the same seed with one variable changed gives a bit-identical twin. On 720 scenes in 12 strata we reimplement seven published seam-extraction methods from their papers and compare, under equal conditions, which geometric mechanism can express which joint. Three findings follow. The plane-intersection mechanism that dominates the literature scores zero on every curved family. The learned segmentation stage that every method depends on, and that no paper releases, accounts for most of every reported score; the one method built on quadric-surface intersection falls from 0.90 to 0.00 median F1 when that stage is withheld. And the hand-annotation protocol the field evaluates against has a floor of 1.4 mm median lateral error on our scenes, $2.4\times$ the accuracy reported against such labels. The generator, the corpus, the reimplementations and the harness are released as an open convention on which methods, including learned ones trained on its rendered tier, can be compared.

I. INTRODUCTION

Welding is among the most widely used joining processes in manufacturing, and its automation with industrial robots has moved from mass production, where a taught program is amortised over thousands of identical parts, toward small batches and structural workpieces whose dimensions, deformations and fit-up differ from part to part [1]–[3]. In that setting the robot cannot be taught by hand for every part: it must perceive the workpiece, locate the seam, and plan the torch path and the tack-weld schedule itself [4], [5]. Three-dimensional vision is the sensing modality this literature has converged on: a depth camera on the wrist, a laser line scanner, or a multi-view reconstruction yields a point cloud of the assembly, and the seam is extracted from it as a curve [1], [2], [6]–[8], for regular workpieces and, increasingly, for irregular ones with curved members and intersecting pipes [2], [7], [9].

Deep learning has entered this perception problem more slowly than one would expect from its role elsewhere in

robot vision. A few point-cloud networks detect seams directly [10]–[12], and active-perception work chooses where to look [13], but most published pipelines keep the geometry classical and use a learned network only for a coarse stage: a weld-region segmentation, a per-surface or per-component mask, a bounding box [1], [2], [6], [7], [14]. We see three reasons in how this literature is evaluated, and this paper addresses each.

The ground truth is not exact. Published methods are scored against a hand-taught path or hand-clicked points in a scan. One recent method [1] states the error budget of its own truth at about 0.5 mm and then reports 0.64 mm RMSE against it. We measured the same annotation protocol with an independent, briefed annotator on scenes whose seams are known exactly and found a median lateral error of 1.4 mm (Sec. VI). A reported 0.6 mm against such labels is a statement about the labels.

The conditions are not varied. Each paper evaluates on its own workpieces, with one sensor, one fixture and, with few exceptions, straight seams. There is a lack of work that reports how a method behaves on a curved seam, with the fixture in view, at twice the sensor noise, or under a different random seed, and the reason is structural rather than a matter of diligence: when the truth is tied to a particular scan, the conditions cannot be varied without re-scanning and re-labelling.

The data are not available, and would not suffice. Datasets are typically “available on reasonable request”. Those that are described comprise a handful of workpieces, one acquisition setup and hand-made labels, which is enough to validate a geometric pipeline on its own parts but is neither large enough, varied enough nor accurately enough labelled to train and compare learning-based perception. A method that others cannot run on the same data under the same conditions has not been compared.

Our answer is one rule: *ground truth is constructed, never detected*. The generator samples a joint (type, thicknesses, root gap, misalignment, included angle, edge preparation), places the parts by closed-form transforms, and the seam is the intersection of the parts’ surfaces computed algebraically from the same transforms. There is no annotator, no estimator and no “consistency” anywhere in the label path, and a corpus of any size and composition can be generated on demand. Everything else follows from the rule: what is stored, what is a versioned convention, and what is a gate.

The contributions are (i) the benchmark: 720 scenes in

12 strata spanning five joint types and six curved seam families, with per-point visibility and reachability flags, a derived stereo sensor model, fixture twins and anti-shortcut gates (Sec. III); (ii) faithful reimplementations of seven published seam-extraction methods, each with its unpublished coarse stage replaced by an oracle from truth so that its geometric mechanism can be measured alone (Sec. IV); (iii) the comparison itself (Sec. V), whose headline is that the dominant mechanism cannot express curvature and that the segmentation stage the papers do not release is most of what they report; (iv) a measurement of the hand-annotation floor of the field’s evaluation protocol (Sec. VI); and (v) the release of the generator, corpus, reimplementations and harness as open source, so that others can generate their own corpora, extend the generator, run their own segmentation stages in place of the oracles, and compare on the rendered tier that the same schema produces.

II. RELATED WORK

Classical seam extraction from point clouds. Before learning entered the pipeline, the seam was found by geometry alone: Zhou et al. [4] reconstruct a workpiece model from partial views, denoise it, and extract the seam from the geometry of the reconstruction; Wang et al. [5], [9] recover annular and curved seams from dense reconstructions and line-laser profiles. These pipelines are the ancestors of the coarse-to-fine designs that follow, and the geometric mechanisms below are what survives of them.

Mechanisms. We group the seven methods we reimplement by the geometric mechanism that produces the seam, because that is what the benchmark measures. *Plane intersection*: fit planes to the workpiece faces (RANSAC [15] in [1], point-pair-feature voting [16] in [6]) and intersect them pairwise. *Crease detection*: find the points where a local descriptor changes, then chain them; the local bounding-box flatness of [2], [17] and the curvature-seeded region growing of [14]. *Slicing*: cut the cloud into slices, take a geometric centre per slice, fit a spline [7]. *Model registration*: register a CAD-like model to the scan and transfer the seam from the model [3]. *Quadric intersection*: fit quadric surfaces to segmented faces and intersect them [8]. Every one of these pipelines relies on a learned segmentation or detection stage in front of the geometry (PointNet++, FastSAM, K-Net, Faster R-CNN, YOLO or DeepLab in the respective papers), trained on data that is not released; the same coarse-to-fine structure appears in neighbouring tasks such as robotic seam coating [18].

Learned seam detectors and datasets. Voxel and point networks detect seams directly [10]–[12], and active-perception work chooses where to look [13]; all are trained and evaluated on scans with hand-made labels and report label *consistency* rather than accuracy. Synthetic data with exact labels is the standard remedy in other perception domains [19], and the value of a shared benchmark with fixed metrics is established [20]; to our knowledge no such benchmark exists for weld seams.

III. THE BENCHMARK

A. Constructed truth

A scene is a set of parametric parts (plates, prisms, prepared plates with ISO 9692-1 grooves [21], tubes with planar or saddle cuts, and swept bands along curves) placed by rigid transforms, and its seams are the analytic intersection curves of their surfaces. For plates the seam is the intersection line of two face planes; for a tube on a plate it is an ellipse from the plane–cylinder intersection; for a tube on a tube a saddle curve; for a swept band an offset of its spine. All are evaluated in closed form and stored as dense polylines (0.1 mm spacing) together with the seam tangent, the two face normals, the local dihedral angle and the torch approach direction. Joint parameters follow the standards: thickness and misalignment ranges from ISO 5817 quality levels [22], root gaps scaling with thickness per ISO 9692-1, and the joint taxonomy of ISO 17659 [23], including the distinction between a T joint and an angle joint that most of the literature elides.

Formally, a part k is a solid whose faces are analytic surfaces $S_{k,i}$ posed by a rigid transform T_k , and a seam candidate between parts A and B is the curve

$$\gamma = T_A S_{A,i} \cap T_B S_{B,j}, \quad (1)$$

kept if the two faces are adjacent within the root gap and the torch cone about the bisector of their outward normals is unobstructed. For two planes γ is a line, for a plane and a cylinder an ellipse, for two cylinders a saddle curve; each is evaluated in closed form, never fitted. A root gap $g > 0$ makes “the seam” ambiguous: the face-plane intersection (nominal), the root line on the far face and the midline of the gap are three curves separated by up to g . We store all three, score against the nominal one, and Sec. VI shows that human annotators click a different one depending on the joint type.

B. What a camera returns

A cloud is sampled uniformly by area on every surface, then flagged twice. `visible_from_cam` is geometric visibility from the stored camera: in the frustum, beyond the sensor’s blind zone, facing the camera and unoccluded by an exact ray cast against the parts. `exterior` is reachability by a perfect multi-view scan, from a 30-direction ray scan. Sensor dropout is deliberately not in either flag: a derived stereo noise model [24] (axial $\sigma_z = sz^2/(fb)$ from a profile’s baseline b , focal length f and subpixel precision s , plus lateral blur and grazing-incidence dropout) is applied at evaluation time from stored parameters, so that noise is a citable convention rather than a baked artefact and can be scaled as an experimental axis. Three sensor profiles ship; the camera is sampled on a spherical shell whose elevation range deliberately includes views in which a standing plate occludes its own seam.

C. Rules, gates and twins

Two more labels are *rules* rather than truth, versioned and re-runnable. The first is the *most probable seam* (MPS). In

practice a welding robot is first positioned coarsely into a weld region and only then asked to find the seam, so its camera is already inside the approach region rather than sampling an arbitrary viewpoint; that workflow is what makes a single-answer task well posed. The MPS of a view v is

$$\text{MPS}(v) = \arg \max_{\gamma \in \mathcal{W}} \int_{\gamma} \mathbb{K}[\text{visible}_v(x)] dx, \quad (2)$$

the weldable seam $\gamma \in \mathcal{W}$ with the greatest visible arclength (ties broken by the fold), evaluated over the exact geometry by a versioned rule. It is a geometric proxy adopted for tractability: which seam carries load is structural knowledge that no single view contains, and we state that scope rather than approximate it. The second rule is a tack-placement schedule (spacing and length as functions of the thinner plate, from shop practice), which yields exact tack positions on every seam. Two corpus-level gates guard against shortcuts: no seam may run parallel to a free plate edge more often than a stated prior allows (outlines are drawn so that the seam-bearing edge is the only axis-aligned one), and the tessellated render meshes must stay within 0.25 mm chord error of the analytic surfaces. Every scene has a *twin key*: the hash of the geometry-determining part of its random state, so that a fixture-present twin, a different sensor profile, or a rendered version of the same scene shares bit-identical workpiece geometry and seam truth. The corpus is a pure function of its configuration and seeds; regenerating it reproduces every content hash.

D. Strata, tasks and metrics

The benchmark corpus has 720 scenes, 60 per stratum, in 12 strata: T joints with a line, a pipe (circle), an obliquely cut pipe (ellipse), a pipe on a pipe (saddle), a rectangular tube (rounded rectangle) and a swept stiffener; corner, lap and edge joints; and square, grooved and arc-shaped butt joints (Fig. 1). *Task 1* recovers every weldable seam from a perfect multi-view exterior scan; *Task 2* names the most probable seam from a single view. Predicted polylines P and true polylines G are densified to 0.5 mm spacing and matched at a tolerance $\tau = 3 \text{ mm}$:

$$\begin{aligned} \text{prec} &= |\{p \in P : d(p, G) \leq \tau\}| / |P|, \\ \text{rec} &= |\{g \in G : d(g, P) \leq \tau\}| / |G|, \end{aligned} \quad (3)$$

with F1 their harmonic mean and $d(\cdot, \cdot)$ the Euclidean distance to the nearest point of the other set; precision penalises phantom seams and recall penalises fragments. We also report the literature’s own matched-path RMSE, the root-mean-square of $d(p, G)$ over matched points. Every method with a random component is run over 30 seeds per scene and reported as the per-scene median; the deterministic methods show exactly zero spread.

E. Tiers

Tier 1, used for every result in this paper, is analytic: geometry, ray-cast visibility and the noise model, with no renderer, no simulator and no GPU, because a generator that needs a specific simulator is as inaccessible as the

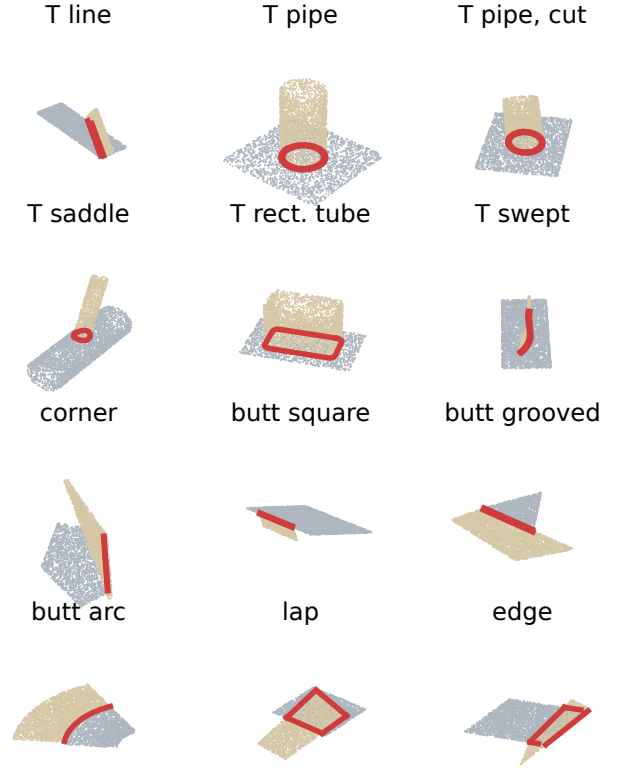


Fig. 1. The twelve strata, one scene each: tier-1 clouds (grey and tan: the two parts) with the constructed seams (red: seams of the joint’s own class; orange: other weldable seams).

data it criticises. Tier 2 renders the same scenes (RGB, depth, validity, seam, tack and object masks) with published material reflectances, CC0 surface sets and workshop HDRIs, paired to tier 1 by seed and gated so that the rendered clean depth back-projects onto the exact surfaces within 0.25 mm ; a 3600-scene training corpus with ten views per scene is being rendered as this paper is written (Fig. 2) and is not evaluated here. The twin gate has already earned its place: comparing the renderer’s exact depth with tier 1’s visibility flags exposed two errors in our own analytic ray test, both corrected and the corpus regenerated before any number here was computed. Two independent exact implementations of one truth disagree only where one is wrong.

IV. SEVEN METHODS, REIMPLEMENTED

Each method was reimplemented from its paper, tested against known geometry, and run through the same harness. Table I lists them with the mechanism and the unpublished coarse stage each depends on. That stage is replaced by an *oracle* from truth of exactly the kind the paper’s network would produce: a weld-region band for M1, per-surface masks for M2, per-component masks for M3, a weld-box crop for M4, per-instance masks for M5, and the welding-surface labels for M7; M6 receives the model itself and is constitutively at the top of the ladder. The oracles are not interchangeable, and supplying the same one to every method

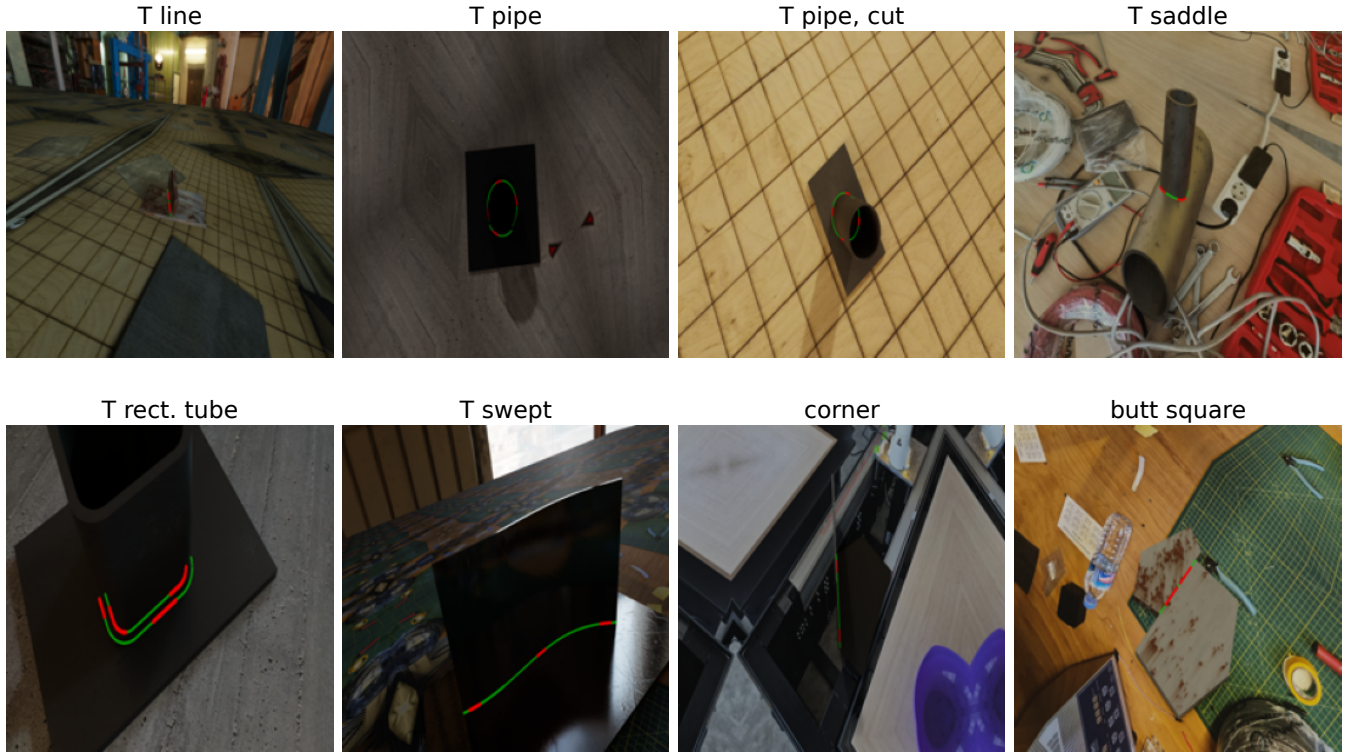


Fig. 2. Tier 2, in progress: rendered views of the same benchmark, with the constructed seam (green) and tack (red) masks. Not used in the results below.

would flatter some and starve others; we measured that a seam band has all of its points within 20 mm of a seam while a surface-derived crop of the same scene has 31%, since a plate has six faces and every rim is a surface junction.

Where a paper reports its own numbers, the reimplementations reproduces them: M1 gives 0.60 mm RMSE and 0.81 mm mean error on a T joint at the paper’s point spacing with its weld region supplied, against reported maxima of 0.64 and 1.16 mm , and exactly zero error without noise and with its refit disabled (three exact points give an exact plane, two exact planes an exact line); M3 gives a median 0.51 mm RMSE on corner joints against a reported $< 0.7\text{ mm}$. Two of M1’s equations are unusable as printed and were implemented as the text intends. The first author of M3 confirmed the implementation choices in writing; correspondence with the authors of M7 is pending.

V. RESULTS

A. Which mechanisms express which joints

Fig. 3(a) is the benchmark’s headline: median F1 per stratum and method under the perfect multi-view scan. Read the rows: a method’s profile across families is its mechanism’s signature.

The plane-intersection collapse is total. M1 holds 0.98 on straight T seams and zero on every curved family and both curved butt strata; M4 degrades more gracefully on curved fillets, since a ring or a saddle is locally a plane pair for short runs, and dies on butts. There is no partial credit at the first non-planar member.

One mechanism survives curvature and one is exact under it. M3’s crease detector is the only method with a non-zero median on every stratum except the grooved butt, at roughly half the accuracy of the leader; M7, the only quadric surface model, is exact wherever its welding surfaces are given (rings and saddles at 0.01 mm) and is the best method on nine of twelve strata.

Grooved butts defeat six of seven methods (median F1 ≤ 0.03); the finding survived rebuilding that stratum with decorrelated outlines, so it is the groove, not the rectangle. Table II gives the straight-to-curved gap per method.

B. The oracle ladder

Fig. 4 withholds each paper’s coarse stage and changes nothing else. The pooled median F1 of M7 falls from 0.90 to 0.00 , of M3 from 0.59 to 0.03 . Two methods that share nothing but that one input fail by the same factor without it, which makes the honest framing a *ladder every method sits on* rather than a weakness of one. M4 has one more rung: with the generator’s exact normals in place of estimated ones its butt and edge scores rise by about 0.5 , so its mechanism is sound and the normal estimate on a thin coplanar pair is what fails.

C. Sensor noise

Fig. 5 scales the derived stereo noise on the single view: clean, the profile’s own σ , and twice it. M7 is essentially noise-immune ($0.52 \rightarrow 0.62$ pooled), M3 degrades, and the plane-intersection methods go to zero at the first noise step.

TABLE I

THE SEVEN METHODS, THEIR MECHANISM, AND THE UNPUBLISHED STAGE EACH DEPENDS ON.

	method	mechanism	coarse stage (oracle)
M1	[1]	multi-plane RANSAC, pairwise intersection	weld-region segmentation
M2	[14]	curvature-seeded region growing, edge points	per-surface masks
M3	[2]	bounding-box flatness, K-means, polynomial fit	per-component masks
M4	[6]	PPF coplanarity, orthogonal-pair voting	weld-box crop
M5	[7]	PCA-adaptive slicing, per-slice centres, B-spline	per-instance masks
M6	[3]	CPD registration [25] of a model; seam transferred	the model
M7	[8]	quadric fits, surface intersection	welding-surface labels

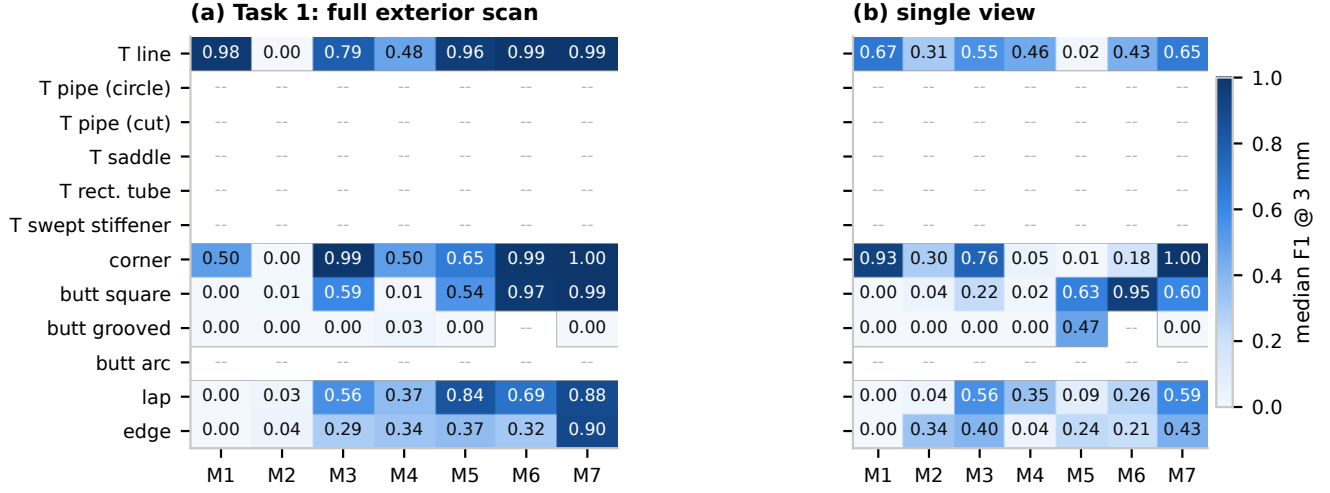


Fig. 3. Median F1 at 3 mm per stratum and method: (a) Task 1, every weldable seam from a perfect exterior scan; (b) the same methods on what one camera returns. M6 registers plate assemblies only.

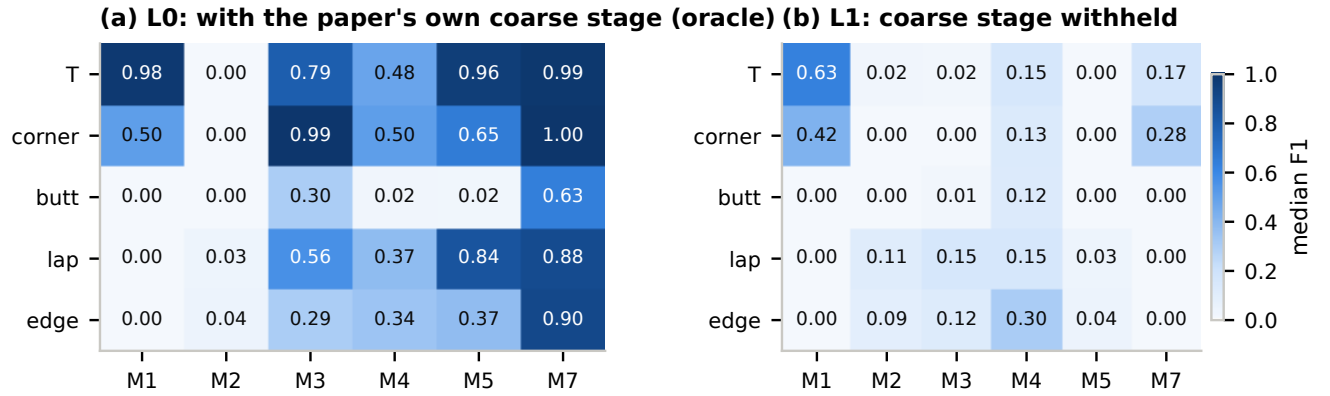


Fig. 4. The oracle ladder: median F1 per joint type (a) with each paper's own coarse stage supplied as an oracle and (b) with it withheld. M6 is constitutively at the top of the ladder and is omitted.

D. Task 2, the fixture, repeatability and cost

Table III scores the most-probable-seam task: whether the named seam received a prediction, and the matched path error. Two methods sit at 100% selection for a reason that is not skill (M5 receives one mask per seam instance, M6 transfers every seam from the model); their localisation is the number to read. M7 localises the named seam within 0.7 mm on every class.

The fixture twins (Table IV) show where a clamping plate in view lands: not on the plane methods, whose band oracle shields them, but on the crease detector and the slicer, which read the fixture-contact lines as seams (M3 adds a median of 23 phantom seams), and on M7, which reads the fixture as a part. Repeatability is measurable here and not reported in the field: a single-draw M1 number is not a measurement, since on a sizeable share of scenes the same cloud gives F1 anywhere from failure to success depending on the RANSAC

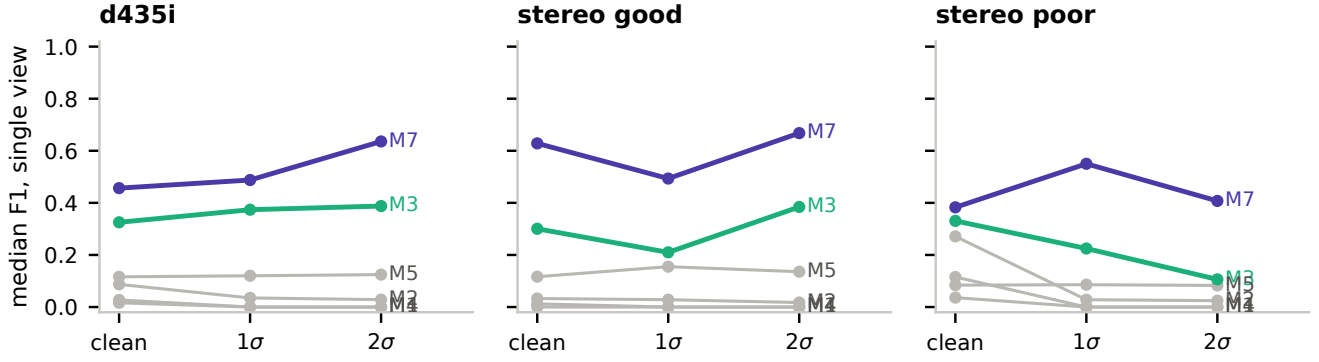


Fig. 5. Median F1 on the single view against the derived stereo noise, per sensor profile. The two leaders are drawn in colour.

TABLE II

STRAIGHT-TO-CURVED GAP: MEDIAN F1 ON THE STRAIGHT T SEAM VS. THE MEAN OVER THE FIVE CURVED T FAMILIES (TASK 1).

	M1	M2	M3	M4	M5	M6	M7
T line	0.98	0.00	0.79	0.48	0.96	0.99	0.99
curved T (mean of 5)	—	—	—	—	—	—	—

seed alone. Every method runs in well under 4 s per scene.

VI. THE ANNOTATION FLOOR

Twenty scenes, four per joint type, were exported as clouds and hand-labelled in CloudCompare by an independent annotator who was briefed with the protocol of the annotation-based literature and did not know the truth. Scored against the exact seams (Fig. 6): the median lateral RMSE is 1.4 mm with a 95th percentile of 11.8 mm; endpoint error is worse still (4.2 mm median); 0.20 of the seams were missed and the flush stack of an edge joint invited marking every visible edge. The lap joint is catastrophic for a human as well (8.5 mm median): the plate edge is picked instead of the toe, the same class that every method above struggles with. Which of the three candidate curves the annotator clicked depends on the joint type exactly as the physical crease predicts: the gap midline on a butt, the root on a fillet. The author’s own pass, excluded from the measurement, was worse (1.9 mm), which supports the exclusion rule rather than undermining it. A sub-millimetre accuracy measured against labels with 1.4 mm of noise is below its own label noise.

VII. DISCUSSION AND LIMITATIONS

What the comparison says. Two findings are structural. The plane-intersection mechanism cannot express curvature, and most of the field’s reported accuracy is the accuracy of a segmentation stage that is not released. The improvement with the highest ceiling is therefore not a new geometric primitive but a welding-surface segmentation that feeds a quadric intersection, and the benchmark makes that claim testable rather than rhetorical. The second structural finding is about evaluation: the protocol the field scores against has

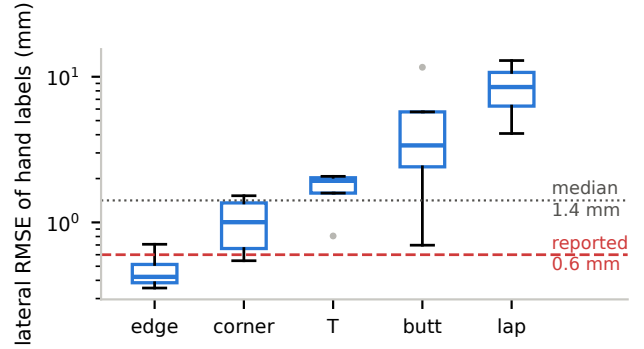


Fig. 6. Lateral RMSE of hand labels against exact seams, per joint type (briefed annotator, 20 scenes), against the accuracy the annotation-based literature reports.

a 1.4 mm floor, and our exact truth is what makes the floor measurable at all.

Limitations. The benchmark is synthetic, and the numbers above are tier-1 numbers. Three answers, stated before a reviewer states them: the tiered design exists precisely because a physical dataset cannot vary these conditions, and the rendered tier and a scanned real subset with known part poses are the next two steps of the same schema; the reimplementations are documented readings with tests and, for one method, the first author’s written endorsement, but they are reimplementations; and the most-probable-seam label is a geometric proxy, a stated convention, not a statement about the load-bearing seam. Our own seam extractor is deliberately not among the seven; the contribution is the dataset and the comparison.

VIII. CONCLUSION

A benchmark whose truth is constructed rather than detected turns three questions the field could not ask into measurements: which mechanisms express which joints, what the unpublished segmentation stage is worth, and how accurate the labels are that published accuracies are measured against. The generator, corpus, reimplementations, harness and results notebook are released with this paper.

TABLE III

TASK 2: SELECTION RATE OF THE MOST PROBABLE SEAM AND MEDIAN RMSE (MM) ON THE MATCHED SEAM, PER JOINT TYPE.

	T	selection rate				T	RMSE (mm)			
		corner	butt	lap	edge		corner	butt	lap	edge
M1	0.93	0.61	0.51	0.36	0.08	0.00	0.00	3.62	0.88	1.96
M2	0.92	0.73	0.88	0.87	0.95	1.35	19.47	19.93	11.35	4.60
M3	0.83	0.35	0.38	0.83	0.95	0.69	0.84	1.24	1.06	1.65
M4	0.62	0.55	0.52	0.73	0.52	0.07	0.24	4.62	3.73	2.26
M5	1.00	1.00	1.00	1.00	1.00	6.10	6.80	2.76	5.31	8.72
M6	1.00	1.00	1.00	1.00	1.00	2.53	4.31	1.49	3.85	4.34
M7	0.73	0.53	0.35	0.80	0.77	0.06	0.02	0.68	0.12	0.29

TABLE IV

THE FIXTURE PRICE: PAIRED TWINS WITH AND WITHOUT THE CLAMPING PLATE IN VIEW (MEDIAN OVER PAIRS).

	pairs	$\Delta F1$	$\Delta precision$	extra seams
M1	40	0.00	0.00	0.00
M2	40	0.00	0.00	2.50
M3	40	-0.07	-0.22	23.00
M4	40	-0.01	-0.01	0.00
M5	40	-0.24	-0.24	-1.00
M6	40	0.01	0.00	-1.00
M7	40	-0.39	-0.51	4.00

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